YUANKAI TAO

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EDUCATION

Georgia Institute of Technology *Aug 2026 — Dec 2027*
M.S. in Operations Research, Stewart School of Industrial and Systems Engineering
Selected Coursework: Deterministic Optimization, Computational Statistics, Simulation

University of California, San Diego *Sep 2024 — Dec 2025*
M.S. in Quantitative Finance, Rady School of Management *GPA:* 3.84/4.0
Selected Coursework: Deep Learning, Computational Finance Methods, Data Science for Finance

Zhongnan University of Economics and Law *Sep 2020 — Jun 2024*
B.S. in Management, School of Accounting and Finance *GPA:* 3.63/4.0
Selected Coursework: Probability & Statistics, Linear Algebra, Calculus, Python Programming

TECHNICAL SKILLS

Programming & Tools Python, SQL, MATLAB, MS Office, LaTeX, Git, Linux, GCP
Quantitative Methods Probability Theory, Simulation, Stochastic Process, Optimization, ML & DL
Libraries & Frameworks Pandas, Polars, Numpy, Scipy, PyTorch, scikit-learn, CVXPy, Backtrader

PROFESSIONAL EXPERIENCE

Guangdong Hedge Fund Investment Management Co., Ltd. *May 2026 — Jul 2026*
Quantitative Research, Intern *Quantitative Strategy Department | Hong Kong*

• Engineered end-to-end commodity futures ML pipeline with 16 features (trend, momentum, volatility, liquidity, volume, open-interest), trained and ensembled LightGBM, XGBoost, CatBoost, and NN across 1- to 20-day horizons, ranked them using rolling out-of-sample validation with both statistical and trading metrics.

• Developed a monthly rebalanced cross-sectional futures strategy using LightGBM return probabilities with 120-day rolling retraining window; implemented in Backtrader achieving 1.89 Sharpe ratio after transaction costs and 15.51% excess return comparing to the benchmark.

• Built a dynamic VaR/CVaR risk measurement system for metal futures; introduced semi-variance slicing method to isolate long/short momentum risks and constructed 17 risk models combining 5 volatility models (GARCH, EWMAV) with 3 distribution assumptions (Normal, StudentSkewed-t, Filtered Historical Simulation).

VariFlight Technology *Jul 2025 — Sep 2025*
Data Analyst, Intern *Data Development Department | Anhui, China*

• Utilized a Python-based stack for web scraping to gather public reports and web pages/images from the Civil Aviation Administration of China (CAAC).

• Employed both traditional and deep learning-based OCR/image recognition tools including pytesseract, Camelot, etc. to extract unstructured data from 600+ non-formatted pdf reports and 500+ webpages and automated the transfer from raw data into formatted csv files as input for further local LLM training.

Ernst & Young *Dec 2022 — Feb 2023*
Assurance, Intern *Financial Service Office (FSO)*

• Assisted the client, Hyundai Motor Finance Company, in auditing financial data related to inter-company and investment transactions; Conducted variance analysis to investigated significant deviations.

SELECTED PROJECTS

Kaggle Jane Street Market Prediction Competition *Silver Medal*

• Built a memory-efficient Polars/PyTorch pipeline for 47M intraday anonymized market observations across 1,669 trading days, engineering lagged, rolling and temporal features, deploying online inference through Kaggle's streaming evaluation API.

• Implemented and evaluated a parameter-efficient TabM regressor, conducting a 32-member shared-parameter

ensemble, and handling with 3 categorical features detected before with one-hot encoding in the training process, obtaining an improvement of +0.004 on CV.

• Contributed to the baseline training and final ensemble of XGBoost, MLP, Ridge, and TabM, increasing the generalization performance of the model sets over fine-tuning, helping the team finish 94th among 3,757 teams.

UCSD / FMSbonds, Inc. 2025 Capstone Project *Aug 2025 — Dec 2025*

Team Lead | High-Performance Bond Pricing Engine Optimization

• Completed the self-defined library by developing a comprehensive municipal bond risk-screening framework covering credit, structural, sector, tax, interest-rate and data integrity risks.

• Designed caching system for high-frequency cash flow queries, eliminating redundant cash flow object generation; reduced per-bond solving time from 1.17 ms to 0.13 ms (9× speedup).

• Rewrote yield-to-worst numerical solver using Numba JIT compilation, reducing computation time from 1.57 ms to 7.5 μs per bond (210× speedup).

RESEARCH EXPERIENCE

Research Assistant for Professor Franklin Qian, UNC-Chapel Hill *Mar 2025 — Jan 2026*

The Numbers Game: Signaling Effects of Pricing Formats in Housing Bargaining

• Engineered large-scale data pipeline (Python, PostgreSQL, PostGIS) processing 76-million-observation spatial dataset on Google Cloud Platform; managed 964.5M-record SQL database with 100GB+ data on UNC Longleaf HPC cluster using Python and DuckDB.

• Implemented parallelized spatial computation workflows to calculate comparable housing variables across multiple distance ranges with property-specific filtering criteria, achieving approximately 50× computational speedup.

• Executed complex geospatial queries to extract population statistics with time-window and property-related constraints.

CERTIFICATIONS & HONORS

Deep Learning Specialization (5-Course Certificate) — Coursera, DeepLearning.AI

Algorithms and Data Structures Certificate (6-Course Certificate) — Coursera, UC San Diego

Beta Gamma Sigma (Top 20% of graduating class) — UCSD, *2024 — 2025*

WorldQuant Alphathon 2024, Gold Level (Rank 16, Top 3.1%) — *2024*

Special Scholarship in Innovation and Entrepreneurship — ZUEL, *2022 — 2023*